

Chapter 6: Euclidean & Coordinate Geometry

Lines, Distance Formulas, Conic Sections & Circle Equations

6.1 Analytic Coordinate Geometry of Lines

Coordinate geometry bridges algebra and geometry by representing geometric shapes on a Cartesian coordinate system.

Fundamental Formulas

Between $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$:

- Distance: $d = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$
- Midpoint: $M = ((x_1 + x_2)/2, (y_1 + y_2)/2)$
- Slope: $m = (y_2 - y_1) / (x_2 - x_1)$

Line Forms & Relationships

- Slope-Intercept: $y = mx + c$
- Point-Slope: $y - y_1 = m(x - x_1)$
- Parallel Lines: $m_1 = m_2$
- Perpendicular Lines: $m_1 \cdot m_2 = -1$

Worked Example 6.1: Perpendicular Bisector

Problem: Find the equation of the perpendicular bisector of segment connecting A(2, 4) and B(6, 8).

Solution:

1. Midpoint: $M = ((2+6)/2, (4+8)/2) = (4, 6)$.
2. Slope of AB: $m_{AB} = (8 - 4)/(6 - 2) = 4/4 = 1$.
3. Perpendicular slope: $m_{\perp} = -1/1 = -1$.
4. Line equation: $y - 6 = -1(x - 4) \Rightarrow y = -x + 4 + 6 \Rightarrow y = -x + 10$.

6.2 Circles & Conic Sections

A conic section is a curve obtained by intersecting a cone with a plane. The four primary curves are the Circle, Parabola, Ellipse, and Hyperbola.

Standard Circle Equation: $(x - h)^2 + (y - k)^2 = r^2$ (Center: (h, k) , Radius: r)
 General Equation: $x^2 + y^2 + 2gx + 2fy + c = 0$ (Center: $(-g, -f)$, $r = \sqrt{g^2 + f^2 - c}$)

Parabola

Vertical: $(x - h)^2 = 4p(y - k)$

Horizontal: $(y - k)^2 = 4p(x - h)$

Vertex at (h, k) , focus distance $|p|$.

Ellipse

Standard: $(x - h)^2/a^2 + (y - k)^2/b^2 = 1$

Foci distance: $c^2 = |a^2 - b^2|$

Eccentricity: $e = c/a < 1$.

Worked Example 6.2: Completing the Square for Circles

Problem: Find the center and radius of $x^2 + y^2 - 6x + 4y - 12 = 0$.

Solution: Group terms and complete the square:

$$(x^2 - 6x + 9) + (y^2 + 4y + 4) = 12 + 9 + 4$$

$$(x - 3)^2 + (y + 2)^2 = 25.$$

Center: $(3, -2)$, **Radius:** $\sqrt{25} = 5$.

Chapter 6 Review & Practice Problems

- Find the perpendicular distance from the point $(3, 2)$ to the line $4x - 3y + 5 = 0$ using $d = |ax_0 + by_0 + c| / \sqrt{a^2 + b^2}$.
- Find the equation of the line passing through $(1, -2)$ parallel to $2x + 5y = 9$.
- Convert the general circle equation $x^2 + y^2 + 8x - 10y + 5 = 0$ to standard form.
- Identify the vertex and focus of the parabola $(x - 2)^2 = 12(y + 1)$.

